

Catch every strip defect where it is born, not where it is shipped

AI surface inspection for stainless strip that detects, classifies and scores every defect at line speed, then routes it to the process that caused it. Built, validated on three public steel datasets, and running live today.

0.764

mAP50 on a held-out test split,
joint model (published: 0.70-0.80)

93.7% to 32.5%

clean steel wrongly flagged,
before vs after clean-strip training

INR 2.5-4.2 cr

per line, estimated, vs USD 1-3M
for a commercial system

~2.2 years

payback for one line at a
conservative 1% downgrade rate



[TRY THE LIVE DEMO](#)

surface-vision.github.io

The trained detector runs in your browser.
Upload a strip image to get box, type,
confidence and severity. Nothing leaves the
page.

1 Problem

Where is value lost?

2 Insights

What did we learn?

3 Solution

What do we build?

4 Implementation

How does JSL deploy it?

5 Impact

What is it worth?

A defect born at the caster is paid for at every later stage, but today it is found at the end of the line

Where stainless surface defects are born, and where they are found without automated inspection

BORN HERE Inclusions, slivers	BORN HERE Rolled-in scale, pits, edge cracks	BORN HERE Pickling patches, over / under-pickle	BORN HERE Scratches, roll marks, crazing	BORN HERE Stains, BA marks, handling scratches	BORN HERE Found here or by the customer
Melt & cast SMS	Hot rolling HSM	Anneal & pickle HRAP	Cold rolling CRM	Final anneal / BA CRAP / BA	Finish & despatch Slit, cut
SEEN TODAY none	SEEN TODAY none	SEEN TODAY spot check	SEEN TODAY spot check	SEEN TODAY visual sample	SEEN TODAY visual + claims

Value added keeps rising left to right: melt, rolling, annealing and pickling are all paid for before the coil is graded. "Seen today" is typical practice without automated surface inspection; JSL's own inspection coverage by line is a PO input.

Cost-of-quality lens (PAF): where AI inspection moves the money

	Prevention	Appraisal	Internal failure	External failure
Today	Process fixes happen after complaints	Manual, sampled, inspector-dependent	Downgrade, rework, re-pickle, scrap	Customer claims, returns, lost orders
With AI	Typed defect sent to its owner in minutes	100% of the surface, the same rule on every coil	Caught upstream, before more value is added	Fewer escapes to exposed-finish customers

Output: shift spend from failure costs (right) to appraisal and prevention (left). Cost of poor quality averages ~15% of sales in manufacturing, range 5-35% (IISE).

Sources: AMETEK Surface Vision, Ternium case study (2020); IISE, Measuring the Cost of Quality; ASTM A240 (304: 8.0-10.5% Ni); LME nickel USD 16,402/t and INR 95.8/USD, 22 Sep 2026; JSL FY26 results.

Why it hurts more on stainless

300 m/min

"...it is impossible for the human eye to inspect the whole surface."
AMETEK / Ternium, cold mill

INR 1.26-1.65 lakh

of nickel in every tonne of 304 (8-10.5% Ni). Downgrading it gives away far more value than on carbon steel

2.57 Mt

JSL FY26 sales at INR 1,67,407/t blended realisation, so every 0.1% downgraded is ~2,570 t

Problem statement

How might JSL detect, classify and score every surface defect at line speed, early enough to act, and tell the right process owner what caused it?

The problem is not that defects exist. They are found late and by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so grade, finish and root cause decide which defect matters

Three facts that change the brief

1 The finish cannot be repaired

For mill finishes like 2B and BA, "surface damage such as scratches, grinding marks or spatter cannot be matched by polishing" (ASSDA).

2 JSL is moving toward surface-critical tonnes

Cold rolling grows 2.05 to 2.67 MTPA by FY28. The 300-series mix held margins in Q1 FY27. Exports are 11% (Japan, Korea, Brazil).

3 End markets see the surface

Automotive, railways, metros and white goods drive demand (Q1 FY27 call). BA goes into appliance interiors.

Framework: grade-by-defect risk matrix (which defects to train for first)

Grade family	Slivers, inclusions	Scale, pickling	Scratches, roll marks	Edge cracks	Ridging, roping	Typical exposed use
200 series (Cr-Mn)	M	M	H	M	L	Kitchenware, tubes
300 series (304, 316)	H	M	H	L	L	Appliances, auto, rail
400 series (409, 430)	H	M	M	M	H	Exhaust, white goods
Duplex (2205)	M	H	M	H	L	Process, infrastructure

H / M / L = relative surface risk. Team hypothesis from metallurgy literature (e.g. TiN-cluster slivers in Ti-stabilised grades, ridging in ferritic 430, poor hot ductility in duplex); to be re-scored in P0 against JSL's defect records. Output: 300 series carries the highest risk on exposed finishes and the most nickel per tonne, so it is the pilot grade.

Framework: root-cause routing (condensed fishbone). Each class the model outputs maps to a process owner

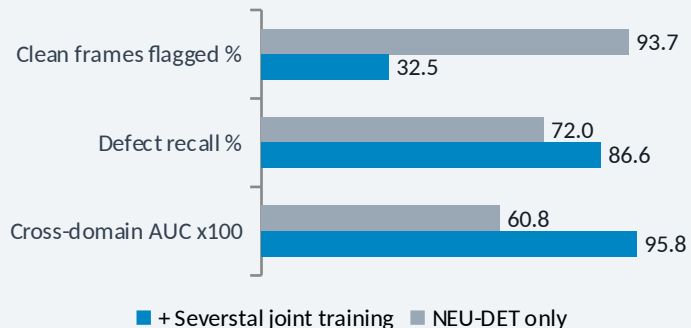
Defect family	Born at	Likely cause (evidence)	Owner	First corrective action
Inclusion, sliver	Caster	Low tundish level, mould-flux entrainment, reoxidation (Leão 2021)	SMS	Hold tundish level, review flux; tag the heat ID
Rolled-in scale, pits	Reheat, HSM	Descaler pressure below spec, blocked nozzles (Steel in Translation 2023)	HSM	Keep descaler above ~180 bar, clear nozzles
Patches, pickling marks	HRAP / CRAP	Uneven scale removal; acid concentration, temperature, line speed	Pickling line	Check acid concentration and speed across width
Scratches, roll marks	CRM, handling	Guides, roller tables, roll surface condition, coil handling	CRM / logistics	Inspect guides, trigger roll change
Edge cracks	HSM	Edge cooling, low hot ductility (duplex, ferritic)	HSM	Edge heating and trim practice; needs edge cameras

Sources: ASSDA, 2B/2D/BA finishes; JSL Q1 FY27 earnings call and results (Aug 2026); JSL cold-rolling expansion to 2.67 MTPA; Steel in Translation 2023 (rolled-in scale); Leão et al. 2021 (tundish slivers). Matrix: team hypothesis.

Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on a 300-series cold-rolled line.

Our Round 1 build produced four findings, and each one changed the design

1 A model that has never seen clean steel raises false alarms on it



On 505 certified-clean Severstal frames the shipped model flagged 93.7%. Training on real clean and defective strip together cut that to 32.5% and raised in-domain mAP50 from 0.752 to 0.764. Training on clean frames alone does not work: recall collapses to 0.057.

So we: train on JSL's own clean and defective strip before any alarm is trusted (P1 gate).

2 Bigger models and higher resolution did not help

Option tested	mAP50	Cost	Verdict
YOLOv8n, 256 px	0.752	3.0 M params	Chosen
YOLOv8n + clean strip (joint)	0.764	same size	Ready
YOLOv8s (3.7x params)	0.734	3.5x FLOPs	No gain
YOLOv8n retrained, 640 px	0.734	4x pixels	No gain
Test-time augmentation	-0.006	2.39x latency	Rejected

Held-out split scored once. Nano vs small 95% CI [-0.008, +0.042] spans zero.

So we: ship a 6.22 MB nano model and spend the budget on coverage, not model size.

3 Throughput, not the model, sets the hardware bill

3,064

inferences per second to see every 0.2 mm pixel of a 1.28 m strip at 250 m/min: about 18 accelerators of the class we measured (range 16-19)

48

per second if each camera frame is downscaled to 1.28 mm per pixel: one accelerator, but fine scratches and crazing fall below one pixel

So we: treat resolution as a P0 specification decision with JSL, and size the edge GPUs from it.

4 Operators only act on alarms they can trust

0.142 to 0.046

calibration error (ECE) after isotonic calibration, down 68%. Before it, a box marked "0.60" was right 84% of the time

0 of 3,200

real steel frames wrongly rejected by our out-of-distribution gate, which blocks non-steel images (the raw model scored a white square as a 0.74 "pit")

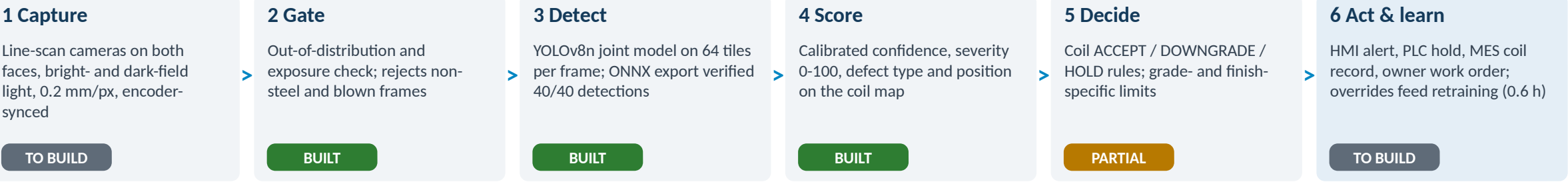
So we: give every alarm a calibrated confidence and a "this is steel" check before it reaches the HMI.

All numbers are measured and reproducible from reports/*.json in our repository: NEU-DET held-out test (180 images), Severstal (505 clean + 191 defective held-out frames), benchmark.json, calibration.json, ood_guard.json.

The model is the cheap part. JSL's own strip, the resolution spec, and trustworthy confidence decide whether the system works.

Detect at the tile, decide at the coil, route to the owner, and learn from every operator override

Architecture: six stages from camera to corrective action



What is new beyond a standard detector

- 1

Clean-strip joint training
Fixes the false-alarm failure that kills pilots. Measured: 93.7% to 32.5%.
- 2

Root-cause routing
Each class maps to an owner (slide 3), so an alarm becomes a work order, not just a red light.
- 3

Unclassified-defect channel
Anomaly model (PatchCore-type) in parallel for defects never labelled. Proposed.
- 4

Learn from JSL's unlabelled coils
Self-supervised pretraining matched ImageNet (0.768 vs 0.773) with zero labels in published steel work.

Framework: build vs buy (options scored against JSL's needs)

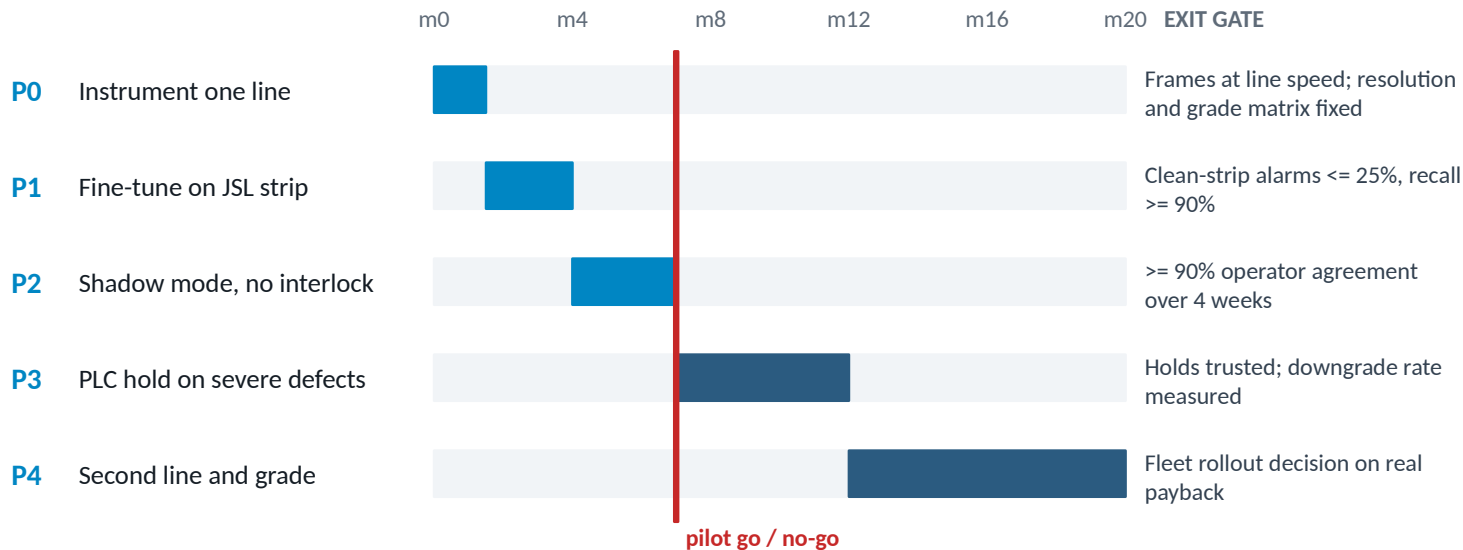
	Capex per line	Proven at line speed	Tuned to JSL grades	Root-cause routing	JSL owns data and model	Time to first line
Commercial (ISRA, AMETEK) USD 1-3M (unverified)						
Fully in-house INR 2-3 cr (est.)						
Hybrid: buy optics, own model INR 2.5-4.2 cr (est.)						
strong partial weak Output: hybrid wins on 4 of 6 criteria and is never weak.						

BUILT = running in our code today (540 tests; live at surface-vision.github.io). Self-supervised result: J. Mater. Inf. 2025 (SimSiam 0.768 vs ImageNet 0.773 mAP50 on steel). Commercial price range: industry, unverified.

Buy proven optics, own the model. JSL gets vendor-grade capture with a detector trained on its own grades and wired to its own process owners.

Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

Stage-gate roadmap (months)



Cost per line, estimated (INR lakh)

Item	Qty	INR lakh
Line-scan cameras and optics	8	60-95
LED lighting, bright + dark field	8	25-50
Edge GPU servers	2-3	45-75
Enclosures, cooling, encoder, network	1 set	30-50
PLC / L2 / MES integration, commissioning	1	60-100
Labelling and fine-tuning (P1)	5,000 frames	30-50
Total capex		250-420
Run cost: 2 engineers, spares, support	per year	~50

Plant integration: read-only first, interlock last

L3 MES / quality	Coil record, disposition, defect map for the customer certificate	P2
L2 Process data	Coil ID, grade, gauge, speed via OPC UA; joins each defect to its heat	P0
L1 PLC	Slow or hold on severe defects; the only write path, enabled in P3	P3

Who does what (RACI)

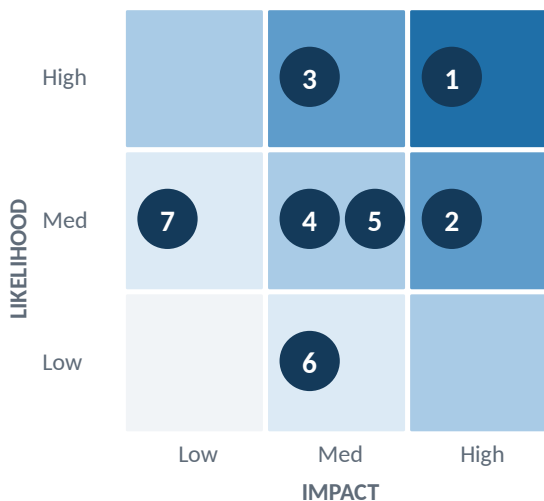
Activity	JSL Quality	JSL Automation	Line ops	Metallurgy	Project team
Install cameras, line access	C	A	R	I	R
Label frames, fix grade matrix	A	I	C	R	R
Train, calibrate, run model	C	C	I	C	A
PLC / MES integration	C	A	C	I	R
Go / no-go at each gate	A	C	C	C	R

Bill of materials is a team estimate at INR 95.8/USD for a 1.3-1.6 m, two-face installation, to be replaced by vendor quotes in P0. Retrain time 0.6 h measured (15.24 s/epoch x 135 epochs).

Commit INR 2.5-4.2 cr and one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

Every high risk is retired in P0-P2, before any interlock can slow the mill

Risk map (likelihood x impact)



Risks, evidence and how each is retired

#	Risk	Evidence	Mitigation	Retired in
1	Model does not transfer to JSL strip	93.7% false alarms before joint training	Fine-tune on JSL clean + defect frames	P1
2	Low-contrast defects missed	Rolled-in scale AP50 0.63	Dark-field lighting; resolution spec	P0-P1
3	Rare classes under-learnt	Roll marks AP50 0.20; no edge-crack data	Edge cameras, active learning, anomaly channel	P1-P3
4	Operators ignore alarms	Uncalibrated scores mislead (ECE 0.142)	Calibrated scores; shadow mode; override log	P2
5	Compute short at line speed	3,064 inferences/s at 250 m/min	Benchmark target GPU on real frames	P0
6	Heat, dust, acid fumes	Pickling and HSM environments	Sealed enclosures, air knives, cooling	P0
7	PLC / MES integration slips	Plant IT change control	Read-only until P3; one write path	P3

Pilot KPIs: how JSL will know it works

$\geq 90\%$

defective frames flagged

measured 92.2% on NEU-DET

$\leq 25\%$

clean-strip alarm rate at P1 exit

measured 32.5% on Severstal

≤ 0.05

calibration error (ECE)

measured 0.046

≤ 200 ms

defect-to-HMI latency

design budget

$\geq 90\%$

operator agreement, shadow mode

measured in P2

< 8 h

retrain for a new line or grade

measured 0.6 h

Evidence column from our measured reports (AP50 per class on NEU-DET and GC10 held-out sets). KPI targets are proposed pilot targets, to be agreed with JSL Quality in P0.

Feasibility is proven where it can be off-site (accuracy, speed, calibration). What only JSL's line can prove is measured in shadow mode first.

At a conservative 1% downgrade rate one line pays back in ~2.2 years; the fleet protects ~INR 13 cr a year

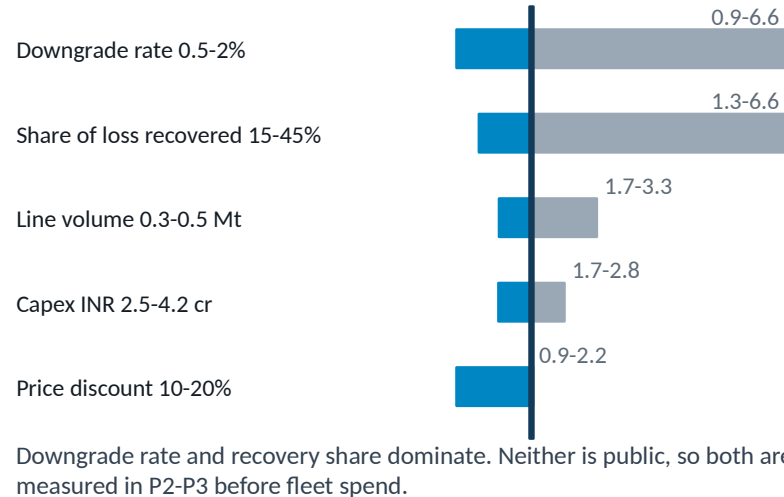
Loss pool per line, INR cr/yr

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times \text{INR } 1,67,407/\text{t}$

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

Outlined: base case, 6.7 cr pool. Recovering 30% = INR 2.0 cr/yr, 1.5 cr net of run cost. Fleet at the same rates (2.57 Mt, ~6.4 lines): INR 43 cr pool, INR 12.9 cr/yr recovered.

What moves payback (years, base 2.2)



Base-case payback

Hybrid (ours)

2.2 yr

Commercial system

12.7 yr

+INR 2.1 cr 5-year NPV per line at 12%

>= 0.11% of line revenue lost to downgrade (rate x discount) is enough for a 2-year payback

Impact beyond the P&L

Coverage

100% of both faces, the same rule on every coil, instead of sampled grading

Customers

Defect map per coil for auto, rail and appliance buyers; fewer claims on exposed finishes

Process

Typed defects tied to heat and coil IDs give the SMS and HSM root-cause data they lack today

Safety and assets

Severe-defect holds guard against strip breaks, the Ternium business case

Sustainability

Less remelt and re-pickling of downgraded coils saves energy, acid and nickel

People

"AI will not replace; it will instead renew and re-energise." Inspectors move to decisions (JSL MD)

WHAT WE ASK OF JSL

- One 300-series cold-rolled line for 7 months
- INR 2.5-4.2 cr pilot budget
- Downgrade rate by line and grade from the MIS
- A metallurgist at 20% to sign off root causes

In return: measured downgrade rate, payback and a go / no-go at month 7

Inputs: JSL FY26 (INR 1,67,407/t; 25,65,902 t). Swept assumptions: 0.4 Mt line, 30% recovered, INR 3.35 cr capex, INR 0.5 cr/yr run cost, 12% rate. Commercial: USD 2M at INR 95.8. MD quote: ET, 16 Jul 2024.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. The number that decides fleet rollout is JSL's own downgrade rate.